



University of Al-Ameed
College of Medicine
Department of Biochemistry



DIGESTION OF CARBOHYDRATES

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Objectives

- ✚ The students will be able to answer questions on the following topics:
- ✚ Digestion of carbohydrate
- ✚ Absorption of glucose
- ✚ glucose transporters

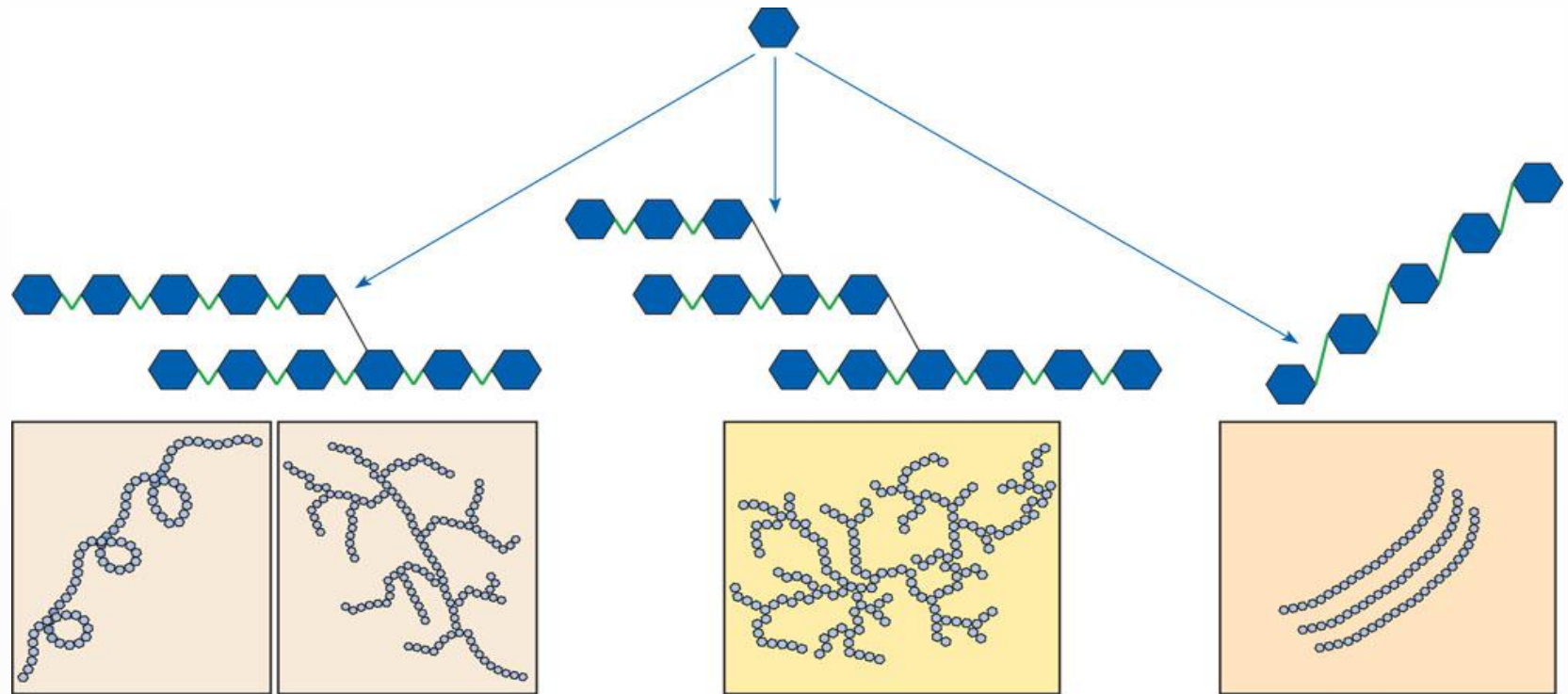
BY studying how the complex carbohydrates present in foodstuffs are broken down to simple sugars in the GI tract.

And studying how the simple sugars are absorbed from the GI tract, into portal blood

Clinical Case Study 1

A newborn baby had severe abdominal distension, severe abdominal pain, and diarrhea after being fed breast milk. Urine analysis revealed the presence of reducing sugar. What is the possible defect? What is the confirmatory diagnostic test? How can it be treated?

Complex Carbohydrates



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Starch

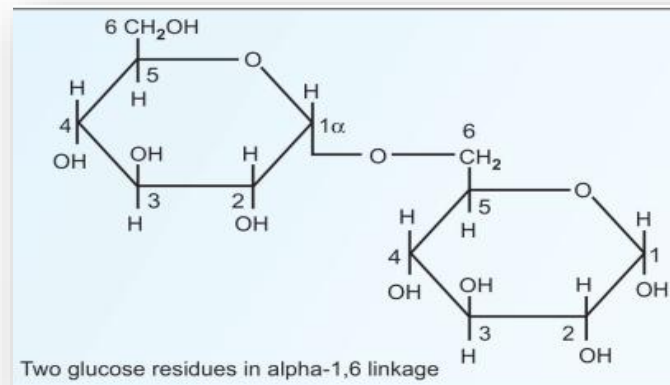
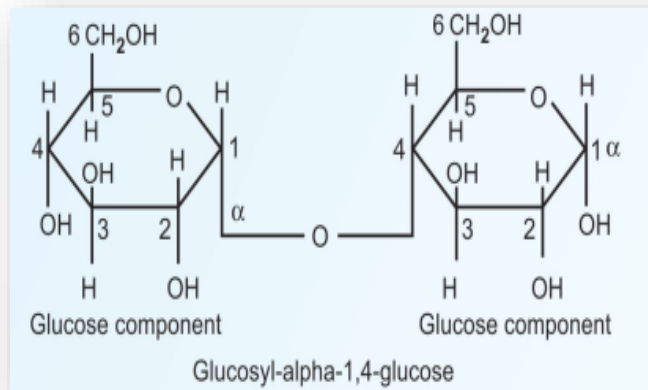
Glycogen

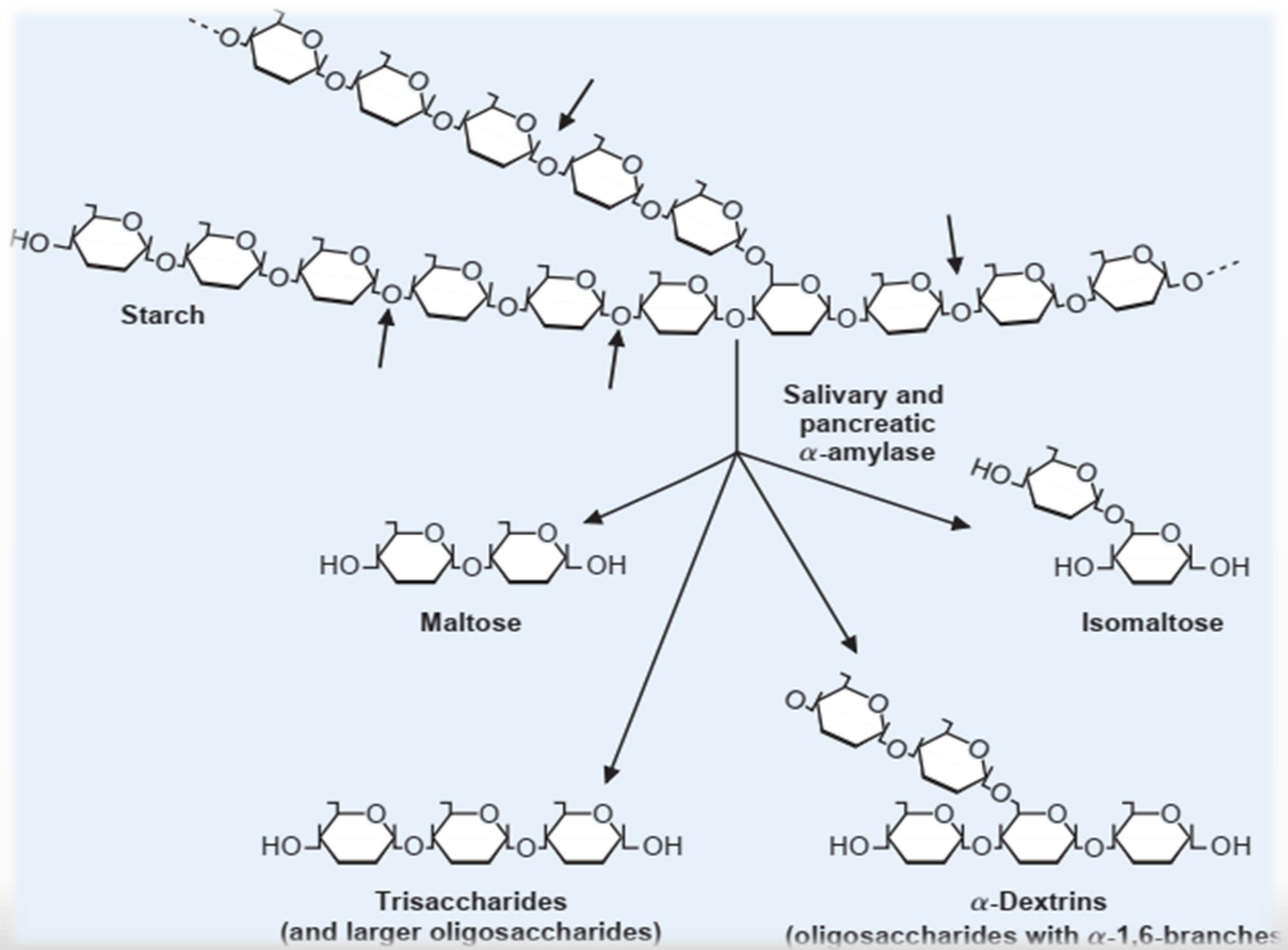
Cellulose

ii. The process of digestion starts in mouth by the **salivary alpha-amylase** which converts starch to smaller polysaccharides called **α -dextrins**.

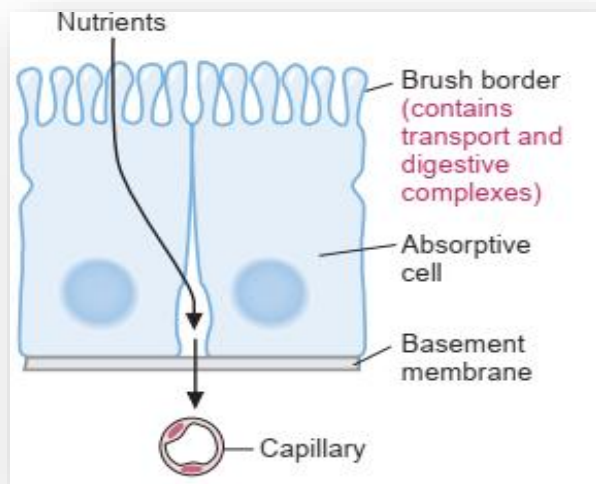
Carbohydrate digestion halts temporarily in the stomach because **Salivary α -amylase is inactivated by the acidity of the stomach (hydrochloric acid [HCl])**

iii. In the **pancreatic juice** another **alpha-amylase** is available which will hydrolyse the alpha-1,4 glycosidic linkages, so as to produce smaller subunits like maltose, isomaltose, maltotriose and branched or unbranched oligosaccharides

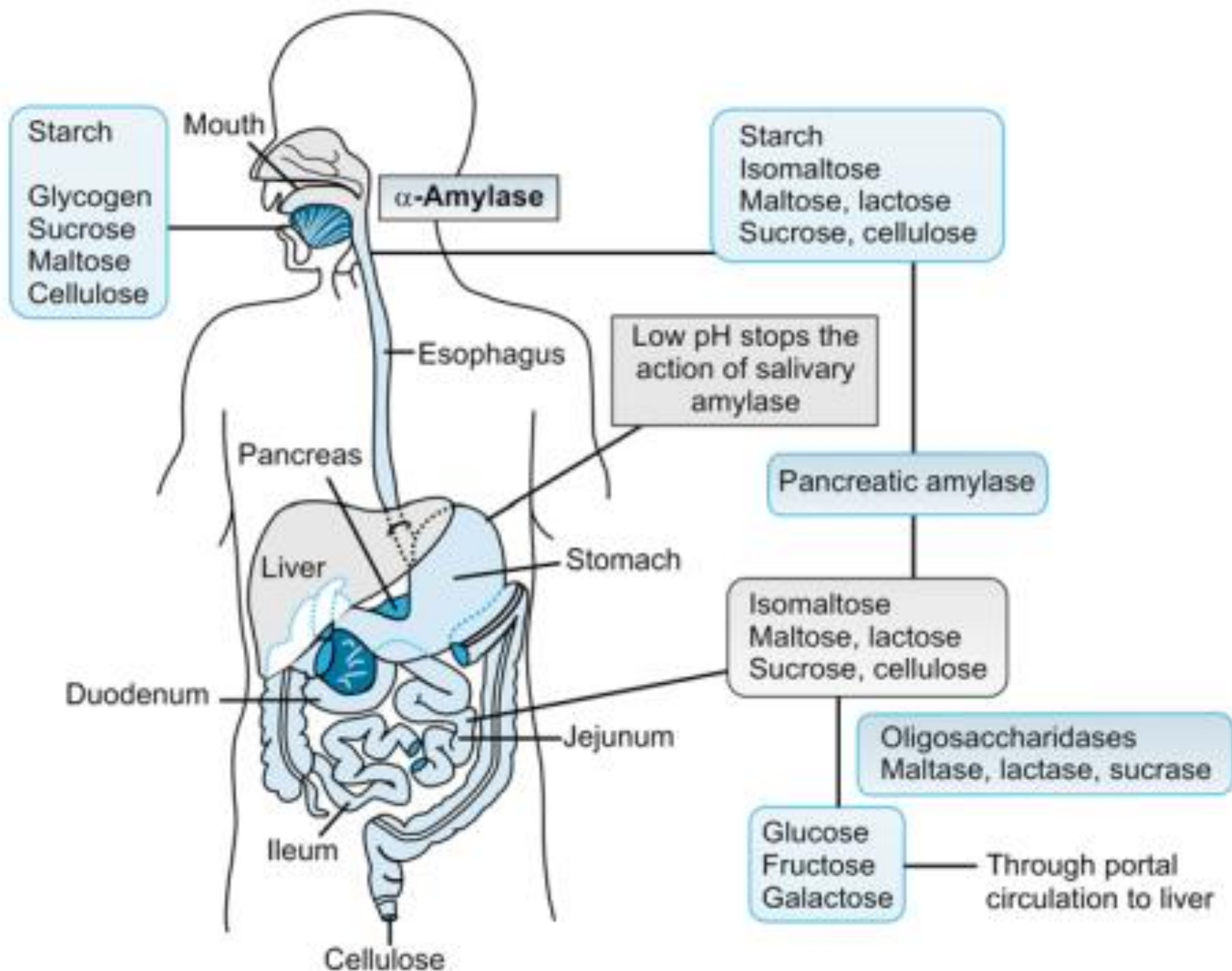




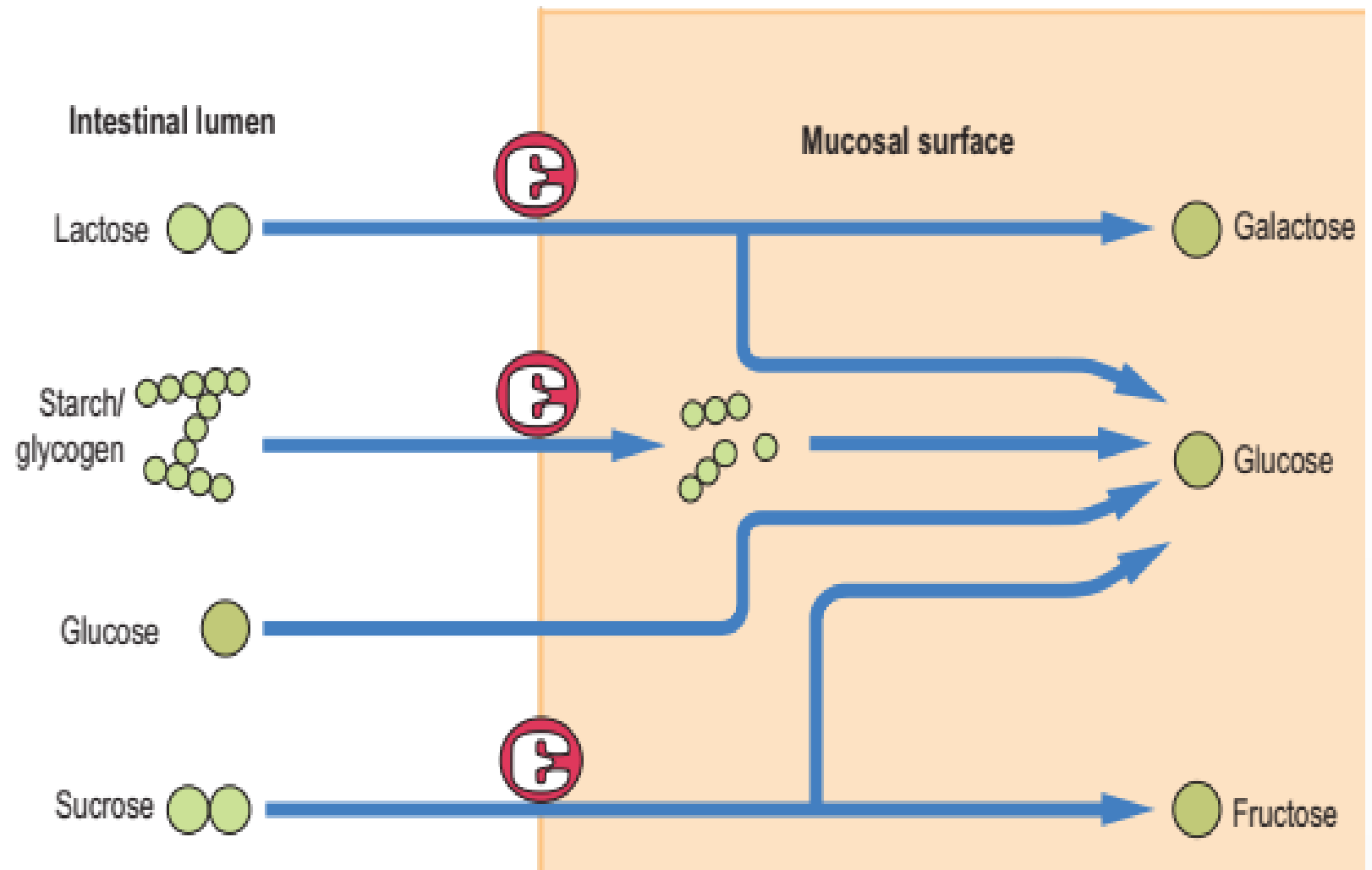
iv. The cells of brush border of intestine contain the enzymes, **sucrase, maltase, isomaltase and lactase**. They hydrolyse the corresponding disaccharides into component monosaccharides which are then absorbed



• Maltose	$\xrightarrow{\text{Maltase}}$	Glucose + Glucose
• Isomaltose	$\xrightarrow{\text{Isomaltase}}$	Glucose + Glucose
• Sucrose	$\xrightarrow{\text{Sucrase}}$	Glucose + Fructose
• Lactose	$\xrightarrow{\text{Lactase}}$	Glucose + Galactose



A Intestinal luminal digestion



Q. Determine the location within the human body where each of the following aspects of carbohydrate digestion occurs.

a. The enzyme sucrase is active.

b. Hydrolysis reactions converting polysaccharides to disaccharides occur.

c. First site where breaking of glycosidic linkages occurs.

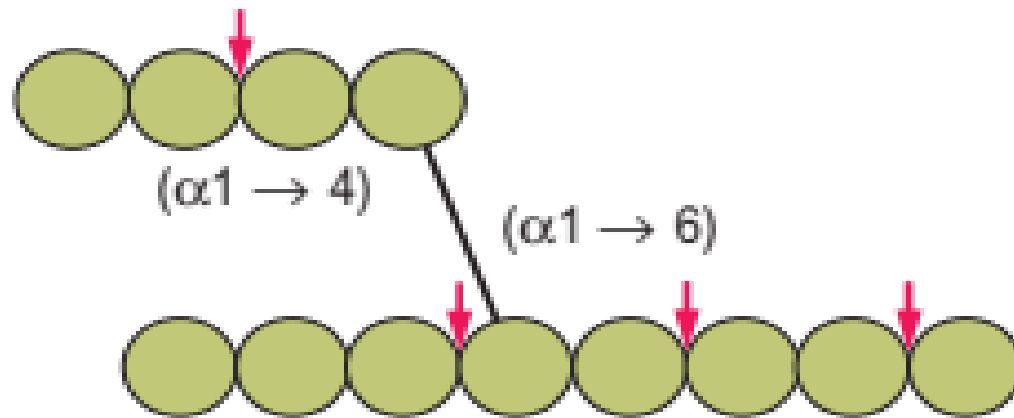
d. The monosaccharides
glucose, fructose, and galactose
are produced.



Q. Determine the location within the human body where each of the following aspects of carbohydrate digestion occurs.

- a. Pancreatic enzymes are active.
- b. Hydrolysis reactions converting disaccharides to monosaccharides occur.
- c. Monosaccharides enter the bloodstream.
- d. The primary site for carbohydrate digestion is located





Hydrolytic cleavage of polysaccharides.



Maltotriose



α -limit dextrin



Maltose



Glucose

Points of
hydrolytic cleavage 

Lactose Intolerance

ii. Deficiency of lactase leads to lactose intolerance. In this condition, lactose accumulates in the gut which undergoes bacterial fermentation in the large intestine with the production of H_2 and CO_2 gases and low molecular weight acids like acetic acid which are osmotically active

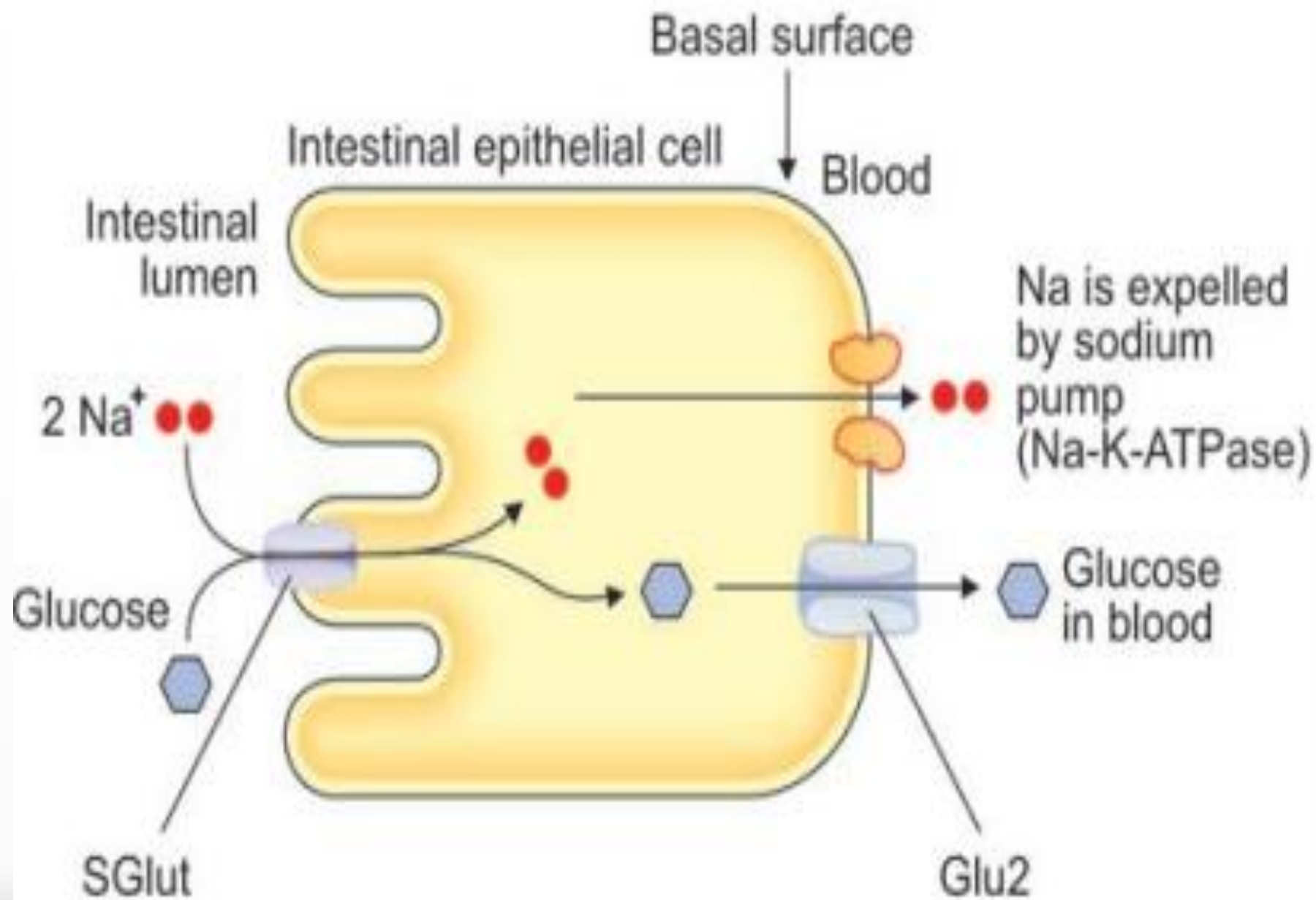
Abdominal cramps and flatulence results from the accumulation of gases and the osmotically active products that draw water from the intestinal cells into the lumen resulting in diarrhea and dehydration

Absorption of Glucose

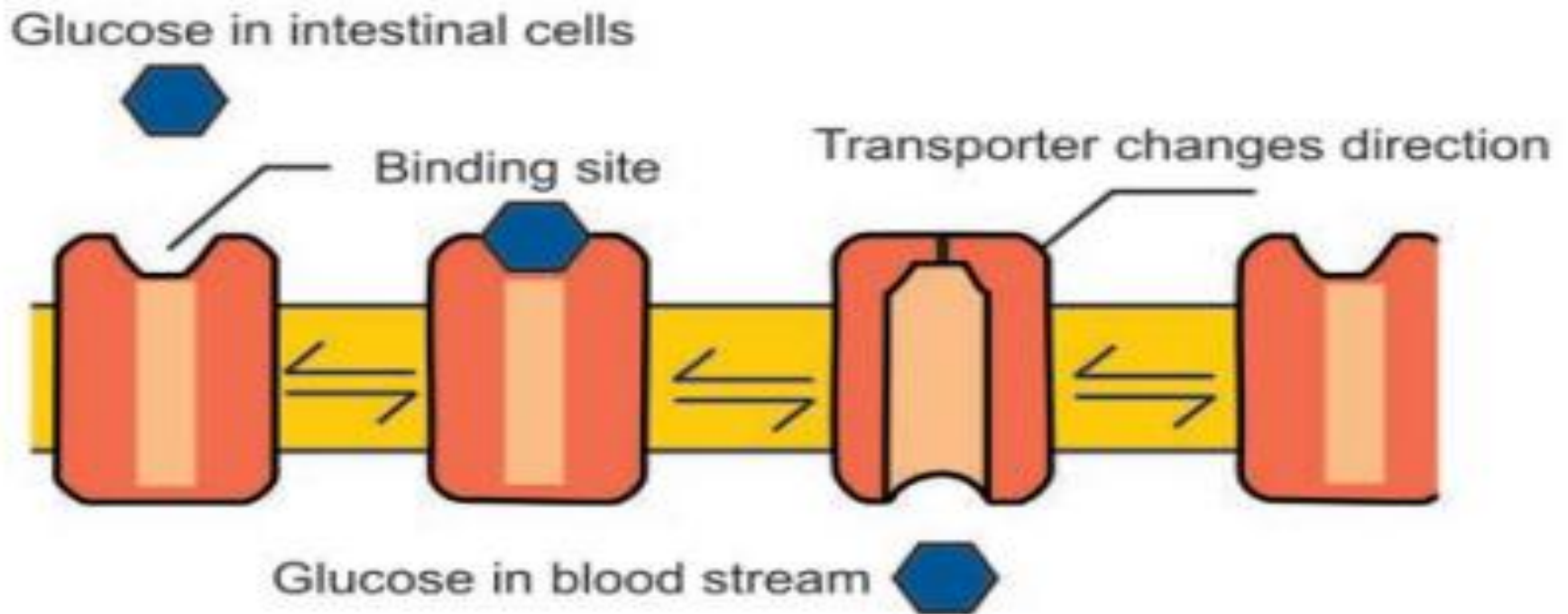
Co-transport from Lumen to Intestinal Cell

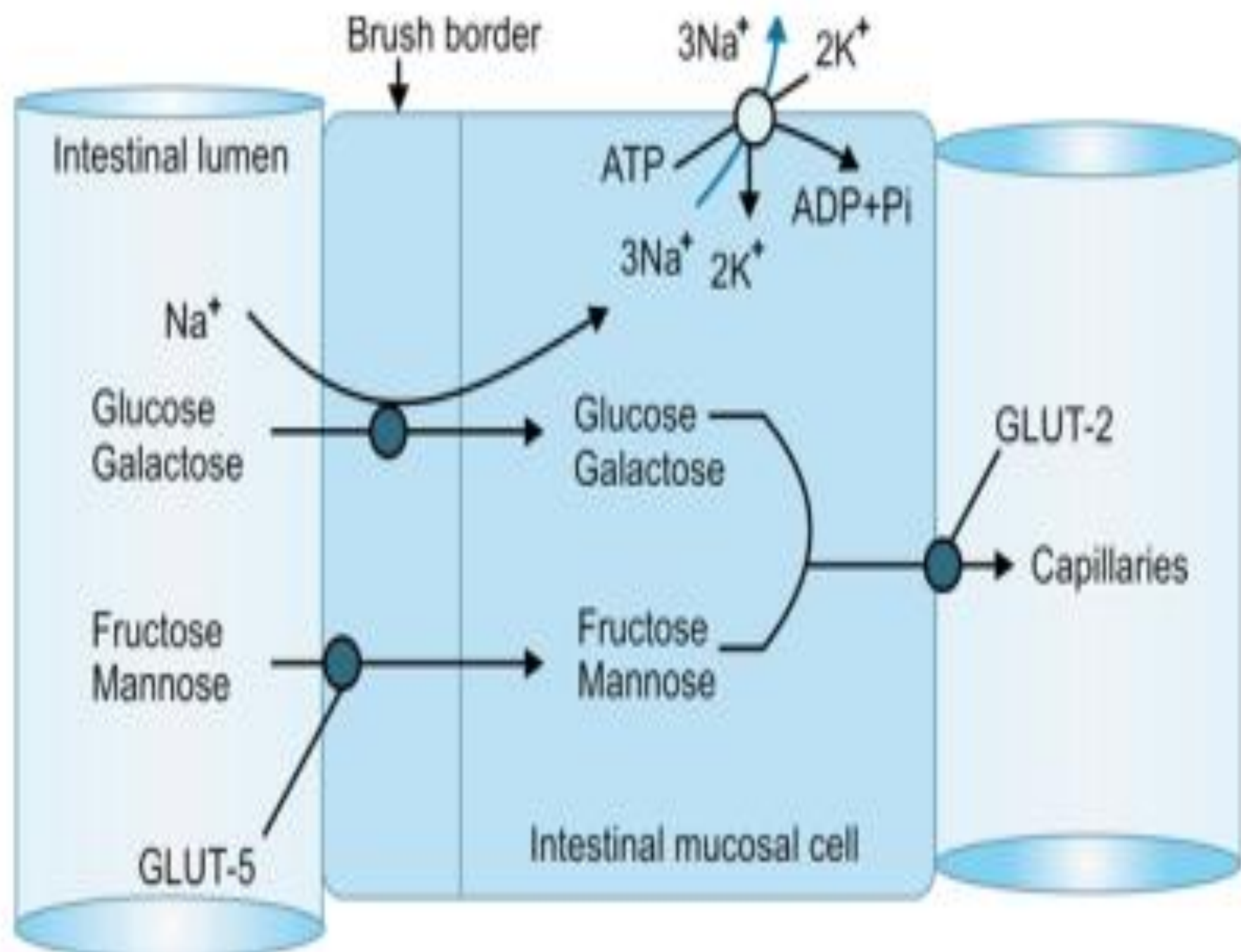
i. This process is mediated by **Sodium Dependent Glucose Transporter-1 (SGlUT-1)**(carrier protein) which carries glucose, along with sodium. This sodium is later expelled by the **sodium pump** with utilization of energy. So energy is needed indirectly (secondary active transport)

Clinical application: Common treatment for diarrhea is **oral rehydration fluid**. It contains glucose and sodium. Presence of glucose in oral rehydration fluid allows uptake of sodium to replenish body sodium chloride.



i. The same intestinal epithelial cells have a different transport mechanism on the membrane facing capillaries. Intestinal cells release glucose into blood stream by the carrier mechanism called **Glucose Transporter Type 2 (GluT2)**. This transporter is not dependent on sodium.





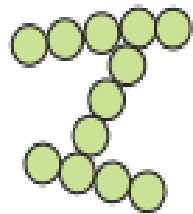
- GluT2 is present in intestinal epithelial cells, liver cells, beta cells of pancreas and kidney
- GluT4 is the major glucose transporter in **skeletal muscle and adipose tissue**

GluT4 is under the control of **insulin**. Insulin induces the movement of intracellular GluT4 molecules to the cell surface and thus increases glucose uptake. In Type 2 **diabetes mellitus**, membrane GluT4 is reduced, leading to insulin resistance in muscle and fat cells.

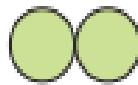
- Very little digestion of carbohydrates occurs in the stomach. Why?
- Where does the digestion of lactose begin, and what is the reaction that occurs?
- Where does carbohydrate digestion begin in the body, and what is the name of the enzyme involved in this initial digestive process?

GluT1	RBC, brain, kidney, colon, retina, placenta	Glucose uptake in most of cells
GluT2	Serosal surface of intestinal cells, liver, beta cells of pancreas	Low affinity; glucose uptake in liver; glucose sensor in beta cells
GluT3	Neurons, brain	High affinity; glucose into brain cells
GluT4	Skeletal, heart muscle, adipose tissue	Insulin mediated glucose uptake
GluT5	Small intestine, testis, sperms, kidney	Fructose transporter; poor ability to transport glucose
GluT7	Liver endoplasmic reticulum	Glucose from ER to cytoplasm
SGluT	Intestine, kidney	Cotransport; from lumen into cell

Carbohydrate (polysaccharide)



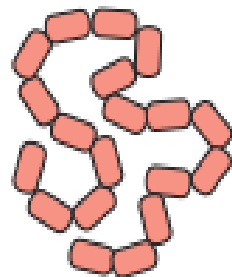
Disaccharide



Monosaccharide



Protein



Peptide



Triacylglycerol (triglyceride)



Free fatty acid



+

Diacylglycerol



+

Monoacylglycerol



+



Thank you

